Mathematical foundations of Bézier's technique

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Bezier's method is one of the most famous in computational geometry. In his book Numerical control *Bezier gives excellent expositions of the mathematical foundations of this method. In this paper a new expression of the functions* $\{f_{n,i}(u)\}$

$$f_{n,i}(u) = 1 - \sum_{p=0}^{i-1} C_n^p u^p (1-u)^{n-p} \quad (i = 1,2,\ldots,n)$$

is obtained.

Using this formula, we have not only derived some properties of the functions $\{f_{n,i}(u)\}$ *(for instance*

$$f_{n,n}(u) < f_{n,n-1}(u) < \ldots < f_{n,1}(u) \quad u \in [0,1]$$

and functions $\{f_{n,i}(u)\}$ *increase strictly at [0,1] etc) but also simplified systematically all the mathematical discussions about Bezier's method.*

Finally we have proved the plotting theorem completely by matrix calculation.

Any parametric curve of degree n was expressed in the form

$$p(u) = \sum_{i=0}^{n} f_{n,i}(u)\mathbf{a}_i \quad (0 \leqslant u \leqslant 1) \tag{1}$$

by Bezier, where

$$f_{n,0}(u) \equiv 1$$

$$f_{n,i}(u) = \frac{(-u)^i}{(i-1)!}\frac{d^{i-1}}{du^{i-1}}\Phi_n(u) \tag{2}$$

and

$$\Phi_n(u) = \frac{(1-u)^n - 1}{u}$$

$\mathbf{a}_0, \mathbf{a}_1, \ldots, \mathbf{a}_n$ are $n+1$ vectors in the space. The vector $\mathbf{a}_0$ points to the beginning point of the curve (Figure 1). The $\mathbf{a}_i$s put end to end form a polygon giving a gross indication of the shape of the corresponding curve. The polygon is called the characteristic polygon or Bezier polygon.

PROPERTIES OF FUNCTIONS $f_{n,i}$

Bezier's curve (1) has many important properties which are the immediate consequences of the properties of the functions $\{f_{n,i}\}$. It is therefore significant to discover the properties of the functions.

Consider $\Phi_n(u)$ as the product of two functions

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Figure 1. Bezier polygon

$$\Phi_n(u) = [(1-u)^n - 1]\,(1/u)$$

Invoking Leibnitz' formula to calculate the derivative of high order, we have

$$\frac{d^{i-1}}{du^{i-1}}\Phi_n(u) = [(1-u)^n - 1]\frac{d^{i-1}}{du^{i-1}}(1/u) + \sum_{p=1}^{i-1} C_{i-1}^p \frac{d^p}{du^p}[(1-u)^n]\frac{d^{i-p-1}}{du^{i-p-1}}(1/u)$$

$$= (-1)^{i-1}(i-1)!u^{-i}[(1-u)^n - 1] + \sum_{p=1}^{i-1}(-1)^{i-1}C_{i-1}^p n(n-1)\ldots(n-p+1) \times (i-p-1)!\, u^{p-i}(1-u)^{n-p}$$

Because

$$C_{i-1}^p n(n-1)\ldots(n-p+1) \times (i-p-1)! = (i-1)!C_n^p$$

we obtain from equation (2)

$$f_{n,i}(u) = 1 - \sum_{p=0}^{i-1} C_n^p u^p (1-u)^{n-p}$$

Setting

$$J_{n,p}(u) = C_n^p u^p (1-u)^{n-p} \quad (p = 0,1,2,\ldots,n)$$

we have

$$f_{n,i}(u) = 1 - \sum_{p=0}^{i-1} J_{n,p}(u) \quad (i = 1,2,\ldots n) \tag{3}$$

It follows at once from (3) that

$$f_{n,i}(u) - f_{n,i+1}(u) = J_{n,i}(u) \tag{4}$$

If we regard $f_{n,n+1}(u)$ as zero, then (4) holds for $i = 0,1,2,\ldots,n$.

Obviously, $u \in (0,1)$ implies $J_{n,i}(u) > 0$, so we have the inequalities

$$f_{n,1}(u) > f_{n,2}(u) > \ldots > f_{n,n}(u) \tag{5}$$

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In particular, since

$$f_{n,1}(u) = 1 - (1-u)^n,$$

and

$$f_{n,n}(u) = u^n$$

we get

$$0 \leqslant u^n \leqslant f_{n,i}(u) \leqslant 1 - (1-u)^n \leqslant 1$$

for $i = 1,2,\ldots,n$ and $u \in [0,1]$. From this it follows that

$$f_{n,i}(0) = 0 \qquad (i = 1,2,\ldots,n)$$
$$f_{n,i}(1) = 1$$

Computing the derivative $f'_{n,i}(u)$ on both sides of formula (2) gives

$$f'_{n,i}(u) = -\frac{i(-u)^{i-1}\,\mathrm{d}^{i-1}}{(i-1)!\;\mathrm{d}u^{i-1}}\Phi_n(u) + \frac{(-u)^i\;\mathrm{d}^i}{(i-1)!\;\mathrm{d}u^i}\Phi_n(u)$$
$$= \frac{i}{u}\;[f_{n,i}(u) - f_{n,i+1}(u)]$$

and according to (4) we find that

$$f'_{n,i}(u) = J_{n,i}(u)\; i/u \;(i = 1,2,\ldots,n) \tag{6}$$

From this we see that for $u \in (0,1)$ $f'_{n,i}(u) > 0$ ie the function $f_{n,i}(u)$ is strictly increasing on $[0,1]$.

The above results can be summarized by saying that the functions $\{f_{n,i}(u)\}$ satisfy inequalities (5) for $u \in (0,1)$ and increase strictly from 0 to 1 on $[0,1]$.

From the point of view of probability, equality (3) indicates that $f_{n,i}(u)$ is the probability of an event A occurring i or more times in n independent tests, where u is the probability of occurrence of the event A in one test. The results are therefore well known in probability theory.

Bernstein's form of Bezier curve

Based on the fundamental formula (3) all the mathematical discussions about Bezier's technique can be systematically simplified. We give an example for illustration.

Let the vectors

$$\mathbf{S}_0 = \mathbf{a}_0$$
$$\mathbf{S}_1 = \mathbf{a}_0 + \mathbf{a}_1$$
$$\mathbf{S}_2 = \mathbf{a}_0 + \mathbf{a}_1 + \mathbf{a}_2$$
$$\cdot$$
$$\cdot$$
$$\cdot$$
$$\mathbf{S}_n = \mathbf{a}_0 + \mathbf{a}_1 + \mathbf{a}_2 + \ldots + \mathbf{a}_n$$

be the $n+1$ vertices of a Bezier polygon. Substituting $\mathbf{S}_i - \mathbf{S}_{i-1}$ for $\mathbf{a}_i$ $(i = 1,2,\ldots,n)$ in (1), we have

$$\mathbf{p}(u) = \mathbf{S}_0 + \sum_{i=1}^{n} f_{n,i}(u)\,(\mathbf{S}_i - \mathbf{S}_{i-1})$$

$$= \sum_{i=0}^{n} [f_{n,i}(u) - f_{n,i+1}(u)]\;\mathbf{S}_i$$

The above expression can be written as

$$\mathbf{p}(u) = \sum_{i=0}^{n} J_{n,i}(u)\;\mathbf{S}_i$$

using (4). (See Forrest[1].) This is the Bernstein form of the curve which connects Bezier curves with the semiclassical theory of Bernstein polynomials, and therefore lays down a solid mathematical foundation for deeper developments of Bezier technique.

Inverse of Associated Matrix

Expanding $(1-u)^n$ involved in $\Phi_n(u)$, gives

$$\Phi_n(u) = \sum_{p=1}^{n} (-1)^p C_n^p\, u^{p-1}$$

and

$$f_{n,i}(u) = \frac{(-1)^i u^i}{(i-1)!} \sum_{p=1}^{n} (-1)^p C_n^p \frac{\mathrm{d}^{i-1}}{\mathrm{d}u^{i-1}}(u^{p-1})$$

by (2). Since for $i > p$

$$\frac{\mathrm{d}^{i-1}}{\mathrm{d}u^{i-1}}(u^{p-1}) = 0$$

then

$$f_{n,i}(u) = \frac{(-1)^i u^i}{(i-1)!} \sum_{p=1}^{n} (-1)^p C_n^p\,(p-1)(p-2)\ldots(p-i+1)u^{p-1}$$

ie

$$f_{n,i}(u) = \sum_{p=i}^{n} (-1)^{i+p} C_n^p C_{p-1}^{i-1} u^p \quad (i = 1,2,\ldots,n)$$

Rewriting the above n equalities as a matrix equality, we get

$$[f_{n,1}(u) f_{n,2}(u)\ldots f_{n,n}(u)] = [u\; u^2 \ldots u^n]\;[M_n]$$

where $[M_n]$ is a lower triangular matrix of order n in which the element in the pth row and ith column is

$$(-1)^{p+i}\, C_n^p C_{p-1}^{i-1}$$

when $p \geqslant i$. $[M_n]$ is called the associated matrix. We are going to find its inverse. After decomposing $[M_n]$ into

$$[M_n] = \begin{bmatrix} C_n^1 & & & \\ & C_n^2 & & \\ & & \ddots & \\ & & & C_n^n \end{bmatrix} [T_n] \tag{7}$$

where $[T_n]$ is the lower triangular matrix in which the element in the pth row and ith column is

$$(-1)^{p+i} C_{p-1}^{i-1}$$

when $p \geqslant i$, the problem of finding $[M_n]^{-1}$ reduces to that of finding $[T_n]^{-1}$.

$[T_n]^{-1}$ is a lower triangular matrix in which the element in the ith row and qth column is

$$C_{i-1}^{q-1}$$

when $i \geqslant q$.

Let us prove this. $[T_n]\;[T_n]^{-1}$ is obviously a lower triangular matrix in which the element in the pth row and qth column is

$$\sum_{i=q}^{p} (-1)^{p+i}\, C_{p-1}^{i-1}\, C_{i-1}^{q-1}$$

when $p \geqslant q$. If $p = q$, the above expression is equal to 1. Let $p > q$. Because of the equality

$$C_{p-1}^{i-1}\, C_{i-1}^{q-1} = C_{p-1}^{q-1}\, C_{p-q}^{p-i}$$

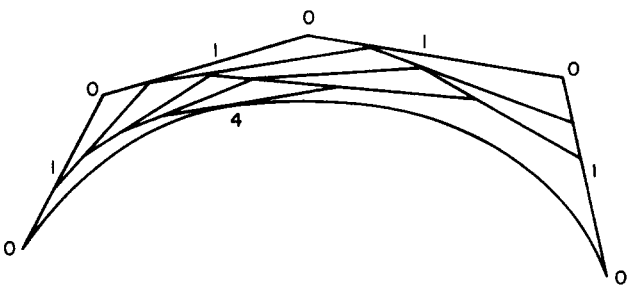

Figure 2. Determining the point corresponding to parameter u *on the curve*

the expression becomes

$$(-1)^p C_{p-1}^{q-1} \sum_{i=q}^{p} (-1)^i C_{p-q}^{p-i} = (-1)^p C_{p-1}^{q-1} \sum_{k=0}^{p-q} (-1)^{p-k} C_{p-q}^{k}$$

$$= C_{p-1}^{q-1} \sum_{k=0}^{p-q} (-1)^k C_{p-q}^{k}$$

$$= C_{p-1}^{q-1} [1 + (-1)]^{p-q} = 0$$

Thus we have proved that $[T_n]^{-1}$ is the inverse of the matrix $[T_n]$. From

$$[M_n]^{-1} = [T_n]^{-1} \begin{bmatrix} 1/C_n^1 & & & \\ & 1/C_n^2 & & \\ & & \ddots & \\ & & & 1/C_n^n \end{bmatrix}$$

we see that $[M_n]^{-1}$ is still a lower triangular matrix in which the element in the ith row and qth column is

$$C_{i-1}^{q-1} / C_n^q$$

when $i \geqslant q$.

PROOF OF THE THEOREM FOR PLOTTING

In his book[2], Bézier suggested an interesting method to determine the point corresponding to the parameter u on the curve (1). Figure 2 illustrates this plotting procedure for $n = 4$ and $u = ¼$.

In all Bézier's works available to us there is no proof of this theorem. Gordon and Riesenfeld proved[3] the theorem based on the Bernstein form of the Bézier curve. We now prove it from the original form (1) of the Bézier curve by matrix computation.

After the first treatment, the $n-1$ sides of the new polygon are the first $n-1$ rows of the n by 1 matrix

$$\begin{bmatrix} 1-u & u & & & & \\ & 1-u & u & & & \\ & & \cdots & \cdots & & \\ & & \cdots & \cdots & & \\ & & \cdots & \cdots & & \\ & & & 1-u & u & \\ & & & & 1-u & \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ \cdot \\ \cdot \\ \cdot \\ a_{n-1} \\ a_n \end{bmatrix}$$

respectively.

We denote the n by n matrix in the above expression by **K**. Similarly after the second treatment, the sides of polygon are the first $n - 2$ rows of the n by 1 matrix

$$\mathbf{K}^2 \begin{bmatrix} a_1 \\ a_2 \\ \cdot \\ \cdot \\ \cdot \\ a_n \end{bmatrix}$$

and so on. Finally, after the $(n-1)$th treatment the single side obtained is the element in the first row of the n by 1 matrix

$$\mathbf{K}^{n-1} \begin{bmatrix} a_1 \\ a_2 \\ \cdot \\ \cdot \\ \cdot \\ a_n \end{bmatrix}$$

Write **K** as

$$\mathbf{K} = u \begin{bmatrix} \lambda & 1 & 0 & \cdots & 0 & 0 \\ 0 & \lambda & 1 & \cdots & 0 & 0 \\ \cdot & & & & & \cdot \\ \cdot & & & & & \cdot \\ \cdot & & & & & \cdot \\ 0 & 0 & 0 & \cdots & \lambda & 1 \\ 0 & 0 & 0 & \cdots & 0 & \lambda \end{bmatrix}$$

where

$$\lambda = \frac{1-u}{u}$$

Set

$$\lambda \mathbf{I} + \mathbf{J} = \begin{bmatrix} \lambda & 1 & & & & \\ & \lambda & 1 & & & \\ & & \ddots & \ddots & & \\ & & & \ddots & \ddots & \\ & & & & \ddots & 1 \\ & & & & & \lambda \end{bmatrix}$$

where **I** denotes the identity matrix of order n and

$$\mathbf{J} = \begin{bmatrix} 0 & 1 & & & \\ & 0 & & & \\ & \cdot & \cdot & \cdot & \\ & \cdot & \cdot & \cdot & \\ & \cdot & \cdot & \cdot & \\ & & & 0 & 1 \\ & & & & 0 \end{bmatrix}$$

Since

$$\mathbf{K}^{n-1} = u^{n-1} (\lambda \mathbf{I} + \mathbf{J})^{n-1}$$

$$= u^{n-1} \sum_{i=0}^{n-1} C_{n-1}^{i} \lambda^{n-i-1} \mathbf{J}^i$$

$$= u^{n-1} \begin{bmatrix} \lambda^{n-1} & C_{n-1}^1 \lambda^{n-2} & \cdot & \cdot & \cdot & C_{n-1}^{n-1} \lambda^0 \\ & \lambda^{n-1} & C_{n-1}^1 \lambda^{n-2} & & & \\ & & \cdot & \cdot & \cdot & \\ & & \cdot & \cdot & \cdot & \\ & & \cdot & \cdot & \cdot & \\ & & & & C_{n-1}^1 \lambda^{n-2} & \\ & & & & & \lambda^{n-1} \end{bmatrix}$$

then

$$\mathbf{S}_0^{(n-1)}\,\mathbf{S}_1^{(n-1)} = u^{n-1}\sum_{i=1}^{n} C_{n-1}^{i-1}\,\lambda^{n-i}\,\mathbf{a}_i$$

$$= \sum_{i=1}^{n} C_{n-1}^{i-1}\,u^{i-1}\,(1-u)^{n-i}\,\mathbf{a}_i \qquad (8)$$

To prove that (8) is tangential to the curve (1) at point $\mathbf{p}(u)$, it is sufficient to prove that (8) is parallel to $\mathbf{p}'(u)$.

$$\mathbf{p}'(u) = \sum_{i=1}^{n} f'_{n,i}(u)\,\mathbf{a}_i$$

and by formula (6)

$$f'_{n,i}(u) = \frac{i}{u}\,J_{n,i}(u) = nC_{n-1}^{i-1}\,u^{i-1}\,(1-u)^{n-i}$$

it follows that

$$\mathbf{p}'(u) = n\sum_{i=1}^{n} C_{n-1}^{i-1}\,u^{i-1}\,(1-u)^{n-i}\,\mathbf{a}_i$$

It is obvious that $\mathbf{p}'(u)$ differs from the righthand side of (8) only by a scalar factor n, so the second conclusion of the theorem follows immediately.

The position vector of the point is

$$\mathbf{a}_0 + u\,\mathbf{S}_0\mathbf{S}_1 + u\,\mathbf{S}_0^{(1)}\mathbf{S}_1^{(1)} + \ldots + u\,\mathbf{S}_0^{(n-1)}\mathbf{S}_1^{(n-1)}$$

$$= \mathbf{a}_0 + u\,[\mathbf{S}_0\mathbf{S}_1 + \mathbf{S}_0^{(1)}\mathbf{S}_1^{(1)} + \ldots + \mathbf{S}_0^{(n-1)}\mathbf{S}_1^{(n-1)}] \qquad (9)$$

$$= \mathbf{a}_0 + [u,0,\ldots,0]\,(\mathbf{I} + \mathbf{K} + \mathbf{K}^2 + \ldots + \mathbf{K}^{n-1}\begin{bmatrix}\mathbf{a}_1\\ \mathbf{a}_2\\ \\ \mathbf{a}_n\end{bmatrix}$$

and it is necessary to compute

$$\mathbf{I} + \mathbf{K} + \mathbf{K}^2 + \ldots + \mathbf{K}^{n-1}$$

first.

Since

$$\mathbf{I} + \mathbf{K} + \mathbf{K}^2 + \ldots + \mathbf{K}^{n-1} = (\mathbf{I} - \mathbf{K})^{-1}\,(\mathbf{I} - \mathbf{K}^n)$$

where

$$(\mathbf{I} - \mathbf{K})^{-1} = \begin{bmatrix} u & -u & & & \\ & u & -u & & \\ & & \cdot & \cdot & \cdot \\ & & \cdot & \cdot & \cdot \\ & & & & -u \\ & & & & u \end{bmatrix}^{-1}$$

$$= \frac{1}{u}\begin{bmatrix} 1 & -1 & & & \\ & 1 & -1 & & \\ & & \cdot & \cdot & \cdot \\ & & \cdot & \cdot & \cdot \\ & & & & -1 \\ & & & & 1 \end{bmatrix}^{-1}$$

$$= \frac{1}{u}\begin{bmatrix} 1 & 1 & \cdots & & 1 \\ 0 & 1 & \cdots & & 1 \\ \cdot & \cdot & \cdot & & \\ 0 & \cdots & 0 & 1 & 1 \\ 0 & \cdots & 0 & 0 & 1 \end{bmatrix}$$

it follows that

$$\mathbf{I} + \mathbf{K} + \ldots + \mathbf{K}^{n-1} = \frac{1}{u}\begin{bmatrix} 1 & 1 & \cdots & 1 \\ & 1 & \cdots & 1 \\ & & \cdots & \\ & & & 1 \end{bmatrix}$$

$$-u^{n-1}\begin{bmatrix} 1 & 1 & \cdots & 1 \\ & 1 & \cdots & 1 \\ & & \cdots & \\ & & & 1 \end{bmatrix}\begin{bmatrix} \lambda^n & C_n^1\lambda^{n-1} & \cdots & C_n^{n-1}\lambda \\ & \lambda^n & \ddots & \vdots \\ & & \ddots & C_n^1\lambda^{n-1} \\ & & & \lambda^n \end{bmatrix}$$

and so the last expression in (9) is

$$a_0 + ([1,1,\ldots,1] - u^n[\lambda^n, \lambda^n + C_n^1\lambda^{n-1}, \ldots,$$

$$\lambda^n + C_n^1\lambda^{n-1} + \ldots + C_n^{n-1}\lambda])\begin{bmatrix}\mathbf{a}_1\\ \mathbf{a}_2\\ \cdot\\ \cdot\\ \cdot\\ \mathbf{a}_n\end{bmatrix}$$

Obviously, the coefficient of $\mathbf{a}_i$ $(1 \leqslant i \leqslant n)$ is

$$1 - u^n\sum_{p=0}^{i-1} C_n^p\lambda^{n-p} = 1 - \sum_{p=0}^{i-1} C_n^p u^n\left(\frac{1-u}{u}\right)^{n-p}$$

$$= 1 - \sum_{p=0}^{i-1} C_n^p u^p\,(1-u)^{n-p}$$

$$= 1 - \sum_{p=0}^{i-1} J_{n,p}(u)$$

ie $f_{n,i}(u)$ by (3). Hence the position vector of the point $\mathbf{S}_0^{(n)}$ is

$$\mathbf{a}_0 + \sum_{i=1}^{n} f_{n,i}(u)\,\mathbf{a}_i = \mathbf{p}(u)$$

Thus we obtain the first conclusion of the theorem and complete the proof of plotting theorem.

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